Small Patients, Big Responsibility: Safe Paediatric Medicine Administration in Pharmacy Practice

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Clinical Reality





When children reject medicines, availability means little.

Overarching aim



Discuss manipulation practices for paediatric medicines and the role pharmacists play in ensuring safe administration of medicines in this population

What paediatric medication administration uncertainties have you experienced in your practice?

More Than a Small Dose: The Challenge of Acceptable Formulations

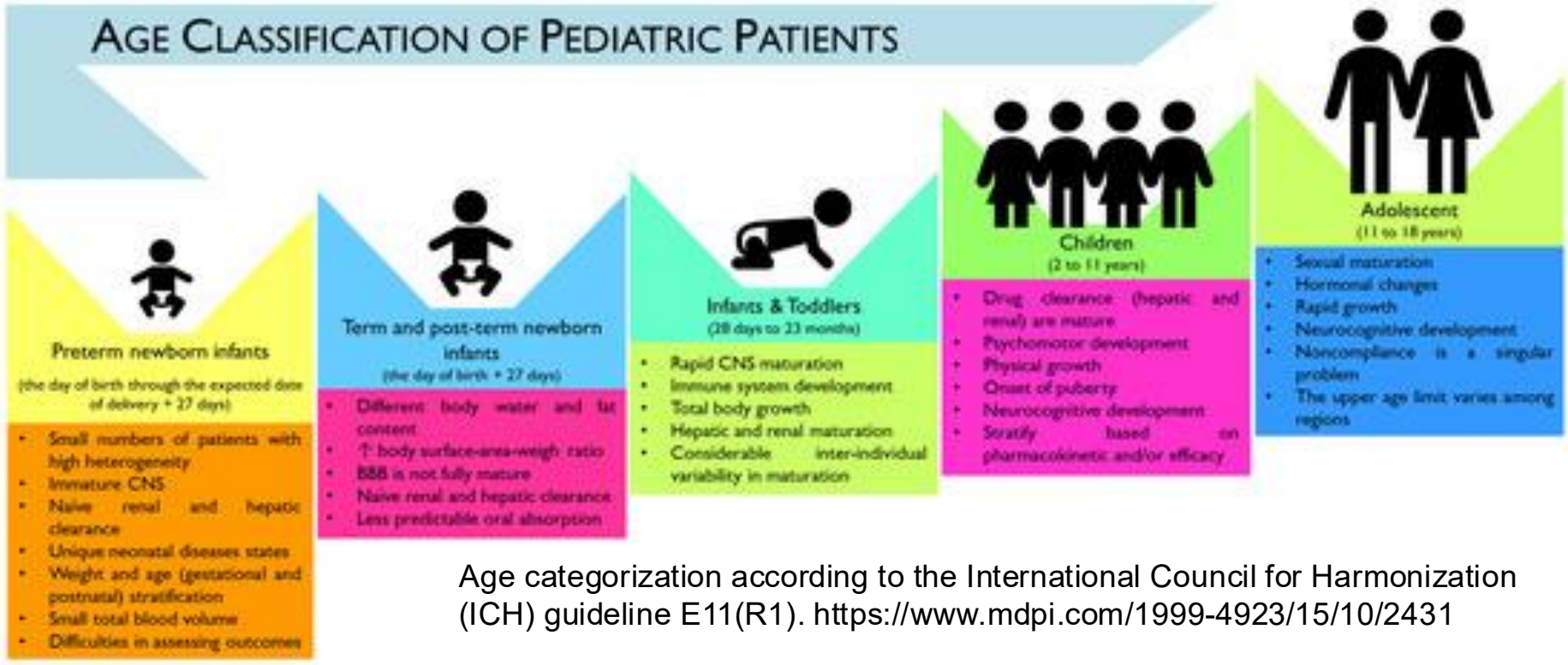
- 5 dimensions of access to medicines: physical availability, affordability, geographical accessibility, quality and acceptability

- Children's medicines should be chewable, palatable, dispersible, crushable and with doses that can be adjusted



Children are not small adults!

Why Children Are at High Risk

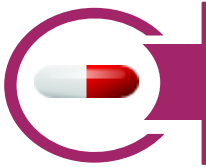


Age categorization according to the International Council for Harmonization (ICH) guideline E11(R1). https://www.mdpi.com/1999-4923/15/10/2431



Children are not Small Adults

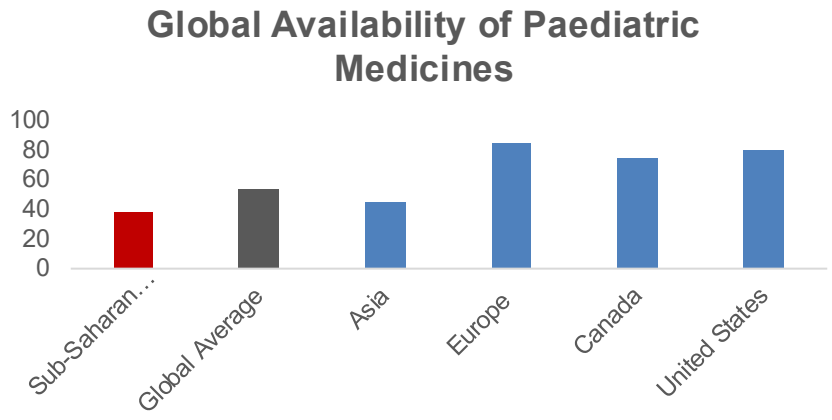
-Small dose errors can cause major harm
-Weight-based dosing increases risk



Limited paediatric formulations

-Dependence on caregivers for administration
-Communication barriers

Global Availability of Child-Specific Medicines



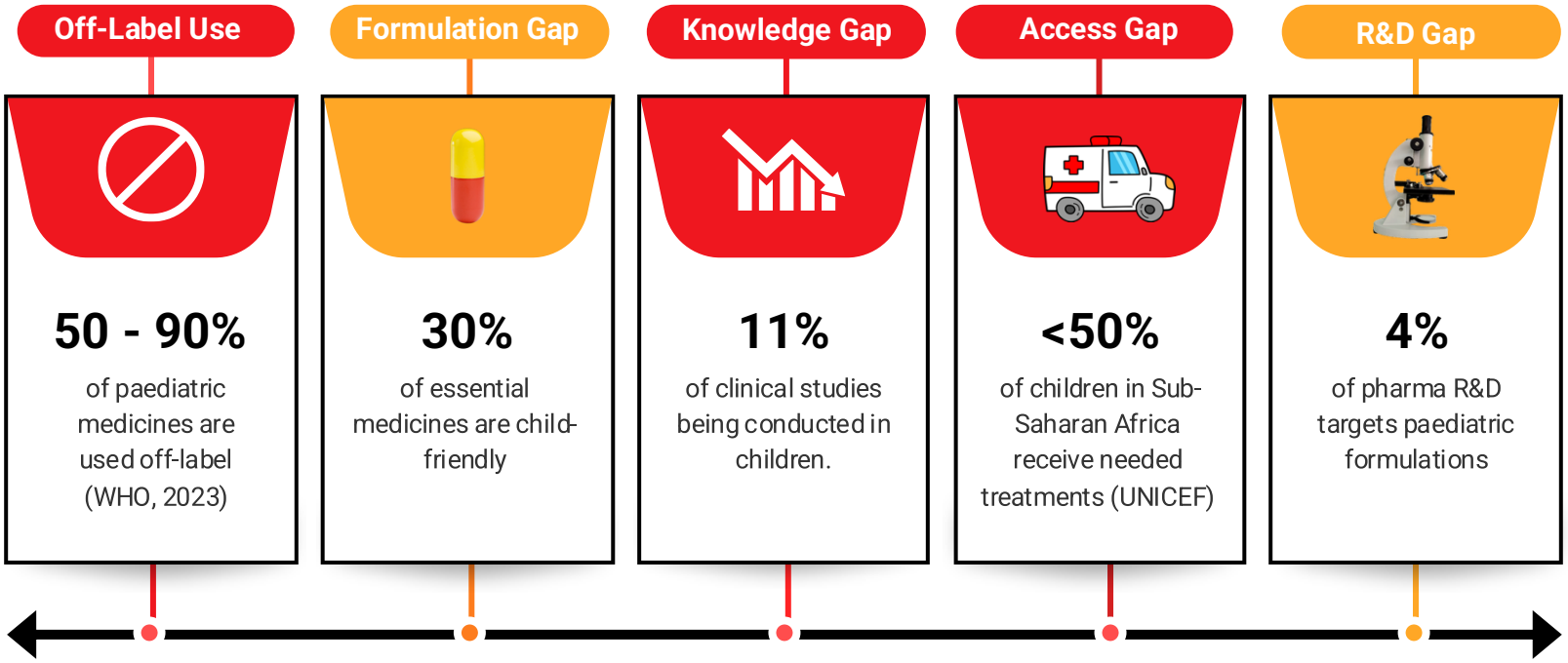


Estimated availability (%) of essential paediatric medicines by region

Sub-Saharan Africa <50% of children receive needed treatments (UNICEF, 2022); Only 30% of essential medicines have child-friendly formulations (WHO Essential Medicines List for Children, 2023).

When children reject medicines, availability means little.

Access to Child-appropriate Medicines



Improvising through manipulating medicines



https://pmc.ncbi.nlm.nih.gov/articles/PMC12420886/

Initiatives from other countries:

- Standardising manipulation practices
- Local specials manufacturers
- Smaller companies specialising in paediatric formulations

In paediatrics, the liquid used and solution made to administer a medicine can determine whether treatment succeeds, fails, or causes harm.

Importance of the Local – What Works for Zimbabwe?

- Set parameters for locally available products
- Design solutions considering locally-available resources and preferences

Testing the stability of isoniazid liquid medicines from a local health facility

- Liquid medicine seemed physically and chemically stable for 7 days (yeast and moulds detected at day 14)
- Positive test for E. coli on day 14 – possible source humans or water.





Before Manipulating Formulations Ask….

1. Can the child really not swallow the tablet/capsule?
2. Is there a licensed liquid available?
3. Is the medicine modified-release or enteric coated?
4. Is crushing contraindicated?



What determines whether we manipulate medicines or not?

Pharmaceutical Considerations in Manipulating Medicines





1. Effect of pH on drug stability



Some medicines are acid-sensitive or alkaline-sensitive.
E.g: Omeprazole/esomeprazole granules

Destroyed by acidic gastric conditions if coating damaged
Mixing incorrectly can reduce effectiveness

The wrong liquid may inactivate the medicine before absorption.

2. Effect of Salt Forms on Solubility

Example Case:

A young child with brain tumour was prescribed oral prochlorperazine mesilate tablets. The drug was discontinued on the market and they resorted to crushing oral tablets.

- Prochlorperazine mesilate (liquid): solubility ≈ 2000 mg/mL
- Prochlorperazine maleate (tablet): solubility ≈ 0.015 mg/mL
- Large difference in solubility of the two salt forms
- Crushed tablets did not provide a reliable dose and led to uncontrolled vomiting

A crushed tablet ≠ commercial liquid preparation

E.g cefuroxime

3. Osmolality and GI Tolerance



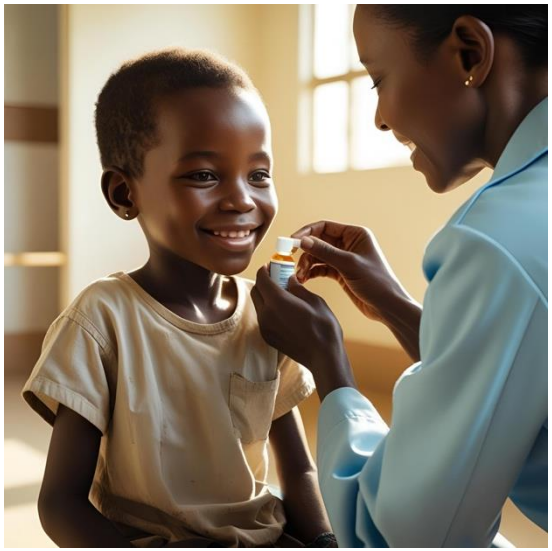
Understand the appropriate concentration for each liquid for all routes of administration

- Some liquids increase osmotic load.
- Risks in neonates/infants diarrhoea, abdominal distension, feeding intolerance.
- E.g. Hyperosmolar syrups diluted incorrectly.

Neonatal concern: premature infants are particularly vulnerable.

Liquids for neonatal use: osmolality of ≤ 450 mOsm/kg to minimize gastrointestinal risks

4. Dose Accuracy



- Mixing medicine into full bottles of milk/juice - unpredictable intake

Medicines should generally be mixed with only a small and/or measured amount of food or liquid.

- Using household teaspoons – inaccurate doses

Use dosing devices, where possible; ensure patient understands how to measure out the medicine



5. Using Appropriate Liquids

Liquid Used	Potential Problem
Milk/Dairy	Reduced absorption of some medicines (ciprofloxacin/iron)
Juice	pH interaction
Tea	Reduced iron absorption
Thick porridge	Incomplete dose delivery
Large feed volumes	Child may not finish dose





Risks of extemporaneous preparations: degradation, microbial contamination, sedimentation, reduced potency.

Conclusions and Call to Action

Safe paediatric medicine administration is not about having perfect systems. It is about pharmacists creating reliable safety checks where children are most vulnerable:

- ward guidelines
- local SOPs
- counselling aids for caregivers
- Initiatives for dosing devices
- administration checklists







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Thank
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